PRAISING HARRIS

QUANTUM MACHINE LEARNING / FAULT-TOLERANT COMPILATION / POST-QUANTUM CRYPTOGRAPHY

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2 upstream PRs (Google Qualtran, PennyLane) · **5,060** trials across 20 SQLite DBs · **58.5%** fault-tolerant T-gate compression · **5.3×** Fubini-Study QNG speedup

PROFILE

Quantum computing & cryptography undergraduate shipping upstream patches to Google Quantum AI (Qualtran PR #1986) and Xanadu (PennyLane PR #10391), and building empirical research engines: Verifiable Fault-Tolerant Quantum Compilation (Kickmix / Gidney MBU / SelectSwap QROM), Fubini-Study QNG, Stim/PyMatching QEC, and NIST FIPS 203/204/205 migration.

EDUCATION

Indian Institute of Information Technology, Kottayam 2024 – 2028 (expected)
B.Tech Computer Science & Engineering (Cybersecurity) | CGPA 7.9
6-Month Co-op Availability: Nov 14, 2026 – Summer 2026 | University NOC Approved & Academic Credits Eligible
Coursework: Quantum Computing, Cryptography, Network Security, Linear Algebra, Algorithms, Operating Systems

PROJECTS

Q-MIND: Autonomous Quantum ML & Verifiable Circuit Compilation Engine github.com/Pyhroff/Q-MIND
Python · PennyLane (PR #10391) · Google Qualtran (PR #1986) · Kickmix SIMD Verifier · 172 tests · 27 300-DPI figures

- Shipped upstream patches to Google Quantum AI's quantumlib/Qualtran (PR #1986: corrected Cuccaro Add.build_call_graph CNOT counts for asymmetric register widths) and Xanadu's PennyLaneAI/pennylane (PR #10391: fixed Evolution PyTree leaves and bind_new_parameters crash for parameter-shift gradients on parametric generators).
- Engineered a Verifiable Fault-Tolerant Quantum Compiler (Gidney Measurement-Based Uncomputation: 4T compute / 0T uncompute + SelectSwap QROM λ=2^k + phase-polynomial fusion) and 64-bit SIMD Kickmix verifier (486M gate-states/s, 100% pass rate), cutting T-count by 58.5% and Toffolis by 48.7% across Fusion 1/√x, DQI GF(2^m), and FeMoCo DF-QROM.
- Derived Fubini-Study Riemannian QNG (5.3× speedup vs Adam; 13.6× barren-plateau gradient recovery; 31.0% REM+ZNE+CDR noise suppression) and validated 5,060 trials across 20 SQLite DBs (classical RBF-SVM beat VQC on 4D tabular data 96.0% vs 91.3%; QCNN and S_N-Equivariant QGNN hit 99.3%–100% on 8-qubit TFIM & SPT phases).

PQC Scanner: Static Detection of Quantum-Vulnerable Cryptography github.com/Pyhroff/pqc-scanner
Python · AST · NIST FIPS 203 / 204 / 205 · pytest · CLI

- Three-bucket taxonomy — QUANTUM-BROKEN (Shor: RSA, ECC, DH, DSA), QUANTUM-WEAKENED (Grover: AES-128 at half effective security) and CLASSICALLY-BROKEN (MD5, SHA-1, DES) — via Python AST analysis plus regex rules for JS, Java, Go and configs (0 false positives on AES-256/SHA-256).
- Recommends ML-KEM-768 for key encapsulation, ML-DSA-65 for signatures and SLH-DSA-128s as a stateless fallback; the ci mode exits 1 on critical findings; HTML, JSON and text reports; 11/11 tests.

QEC Lab: Surface-Code Simulator github.com/Pyhroff/qec-lab
Python · Stim · PyMatching v2 · FastAPI · D3.js

- Rotated surface-code simulator with circuit-level noise (Stim + PyMatching v2); the threshold curve for d = 3, 5, 7 converges at p ≈ 1%, consistent with Fowler et al. 2012.
- Lattice-surgery X-merge of two patches at a rough boundary, decoded with MWPM, plus an RSA-2048 qubit estimator that tracks Gidney's 2019 (~20M qubits) and 2025 (under 1M) results.

Quantum Collapse: CRQC Infrastructure Cascade Simulator github.com/Pyhroff/quantum-collapse
Python · NetworkX · FastAPI · D3.js

- Mosca's-inequality calculator for the harvest-now-decrypt-later window, with a 17-node dependency graph (central banks, certificate authorities, cloud PKI, government systems) and weighted trust propagation.
- A root-CA compromise cascade-fails 11 of 17 nodes; an interactive slider shows the PQC adoption rate where early migration halves the blast radius.

ZeroTrust-AI: Zero-Knowledge Proofs for AI Accountability github.com/Pyhroff/zerotrust-ai
Python · Schnorr · Fiat-Shamir · pytest

- Implemented a Schnorr proof from scratch (key generation, commitment R = kG, challenge c = H(R || message), response s = k + cx mod q, verification) with no crypto libraries in the core, made non-interactive with Fiat-Shamir.
- Built the anonymous prompt-evidence protocol: proves a prompt was submitted and logged without revealing its content or the submitter. Two further protocols (hidden-suite safety auditing, verifiable unlearning) are specified as designs; the AI-model component is a stub.

SKILLS

Cryptography Schnorr and sigma protocols, Fiat-Shamir, commitment schemes, anonymous credentials, NIST FIPS 203 (ML-KEM), 204 (ML-DSA), 205 (SLH-DSA), Mosca's inequality, CRQC threat modeling

Quantum ML & Compilation Verifiable Kickmix compilation (Gidney MBU, SelectSwap QROM), Fubini-Study QNG, barren plateaus, ZNE/CDR/REM mitigation, QCNN, S_N-QGNN, rotated surface codes (Stim, PyMatching v2), PennyLane, Qiskit

Engineering Python, SQL, C/C++, Bash; NumPy, SciPy, Scikit-learn, SQLite, FastAPI, NetworkX, D3.js, pytest, GitHub Actions, Docker

CERTIFICATIONS & WRITING

Training TryHackMe AI Security path (Jun 2026) and Pre-Security path, top 2% globally. Fortinet: Network Security Expert Level 1 (NSE 1). Medium: six security articles.